CUSTOMER RELATIONSHIP MANAGEMENT (TYPE B)

MARKETING ANALYTICS
TOPIC 3
A.A 2024/2025

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1. INTRODUCTION

OVERVIEW

In a competitive retail landscape, understanding customer behavior is essential for driving loyalty and profitability. This project, focuses on enhancing Customer Relationship Management (CRM) through data analysis. By leveraging transactional and customer data, the project aims to provide actionable insights to improve retention, reduce churn, and optimize marketing strategies.

Objectives

The project aims to leverage data-driven approaches to segment customers, identify those at risk of churn, and discover purchasing patterns. These insights support CRM strategies to improve retention and maximize sales.

Datasets

Two datasets were analyzed: Dataset A, focused on transactional details for Market Basket Analysis, and Dataset B, centered on customer transactions for RFM and Churn analysis.

Relevance of CRM Optimization

Effective CRM helps retain customers, reduces acquisition costs, and increases lifetime value. Data analytics enables personalized strategies, improving loyalty and profitability.

2. DATASET PRESENTATION AND ANALYSIS

DATASET OVERVIEW

DATASET A

- **Observations**: 1,093,471
- **Variables**: 13
- **Focus**: Market Basket Analysis (MBA) to identify product associations and cross-selling opportunities

DATASET B

- **Observations**: 2,071,167
- **Variables**: 10
- **Focus**: RFM and Churn Analysis to segment customers and predict loyalty risks

DATASET CLEANING

DATASET A

- **Handling Missing Values:** replaced "NULL" with NA across all columns
- **Inconsistent Rows Removal**: removed rows where UNITS = 0 but price.per.unit was not NA, and removed no sense negatives or 0
- **Imputation of Missing Values**: filled missing BRAND and SUB.BRAND using unique values from Item_id, defaulting to "NO BRAND" when necessary.
- **Calculated Total_Expense** based on available price and quantity data.
- **Receipts Table**: grouped data by Customer_id and DATE to compute daily expenses (Total_Receipt_Expense) and transaction counts (Number_of_Transactions).
- **Customer ID Validation**: identified suspicious Customer_id entries (e.g., incorrect length, invalid characters).

DATASET B

- **Data Integrity Checks**: Filtered the data to ensure that there are no inconsistencies such as sales records with zero items bought or where PL_gross_sales doesn't match the number of items. Rows where no items were bought (number_items_other + number_item_PL == 0) are filtered out. Checked for negative or zero values in relevant columns and addressed them.
- **Sales Data Cleaning**: Gross Sales Validations: Rows where net_sales > gross_sales or PL_gross_sales > gross_sales are removed.
- **Thresholds**: Records with net_sales < 0.1 are removed.
- **Sales Difference**: A new column diff_sales is created to store the absolute difference between net_sales and gross_sales. The distribution of this difference is visualized, and extreme values are flagged.

2. RFM- METHODOLOGY

Objectives

The RFM analysis was conducted to segment customers based on their purchasing behavior and provide actionable insights for personalized marketing strategies

Dataset preparation & Metrics

The dataset was further cleaned and checked to **remove missing or anomalous values**, such as negative sales. In addition, a group by for id_customer and date has been created, so that daily "Receipts" can be created

BASIC METRICS COMPUTATION

- **Recency**: Measures the number of days since the customer's last purchase, and it is calculated as Recency=Today's Date−Last Purchase Date. In this way it is possible to identifies how recently a customer engaged with the business, crucial for targeting active vs. inactive customers
- **Frequency**: Counts the total number of purchases made by each customer within a defined period and is calculated summing the number of transactions for each unique customer ID making it possible to divide highly engaged customers from occasional buyers.
- **Monetary**: Captures the total value of purchases made by each customer and is like Monetary=∑(Transaction Amounts) per customer

RFM Segmentation

(0-5 Days: High, 5-12 Days Medium, 12+ Days Low)
For all the metrics, customers were grouped into three categories according to the following criteria:

Recency (R): A first attempt with percentiles resulted in less realistic segmentation, thus the latter was done according to the number of days

- High (H): Recency lower than 5 days (most recent buyers)
- Medium (M): Recency between 6 and 12 days
- Low (L): Recency higher than 12 days (least recent buyers)

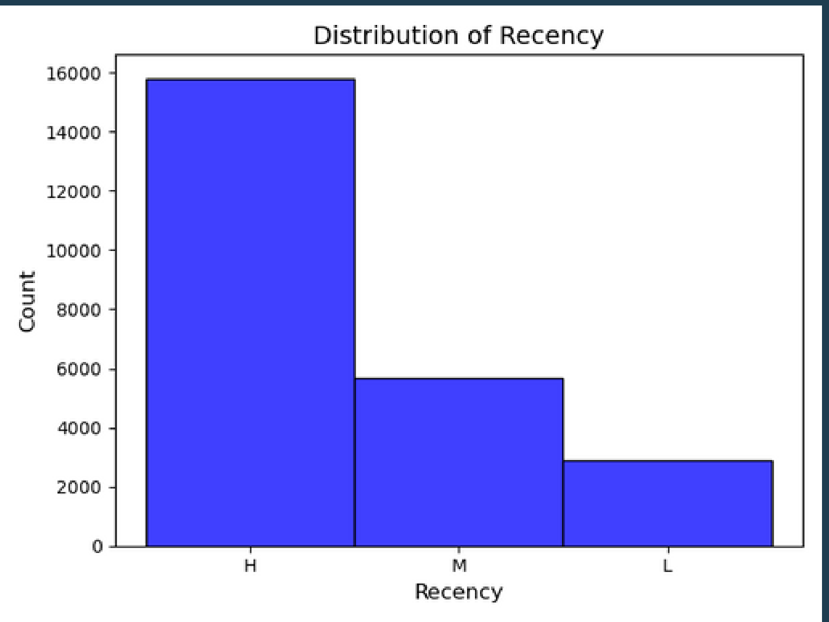
Frequency (F) and **Monetary (M):** For both metrics we used quantiles(), in particular:

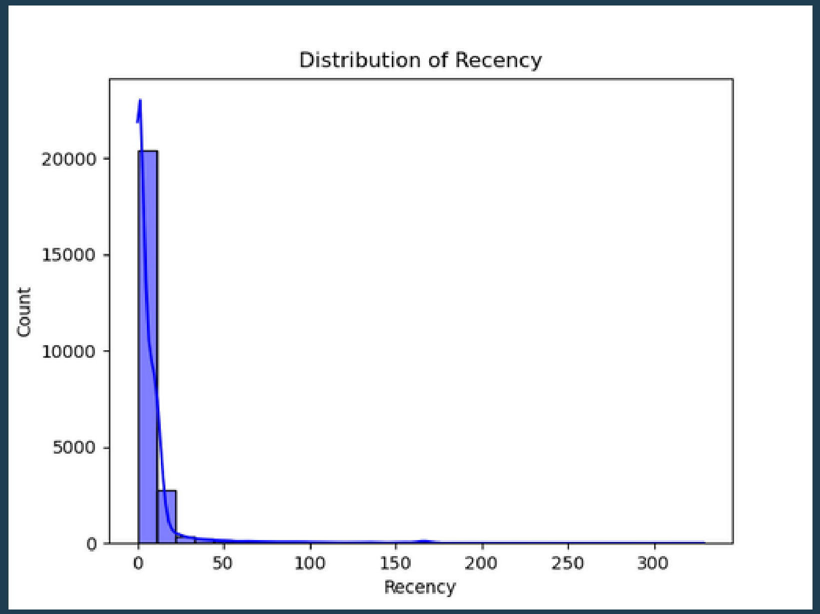
- High (H): Customers in the top third (frequent/top spending buyers)(value>p80)
- Medium (M): Customers in the middle third (p50<value<p80)
- Low (L): Customers in the bottom third(value<p50)

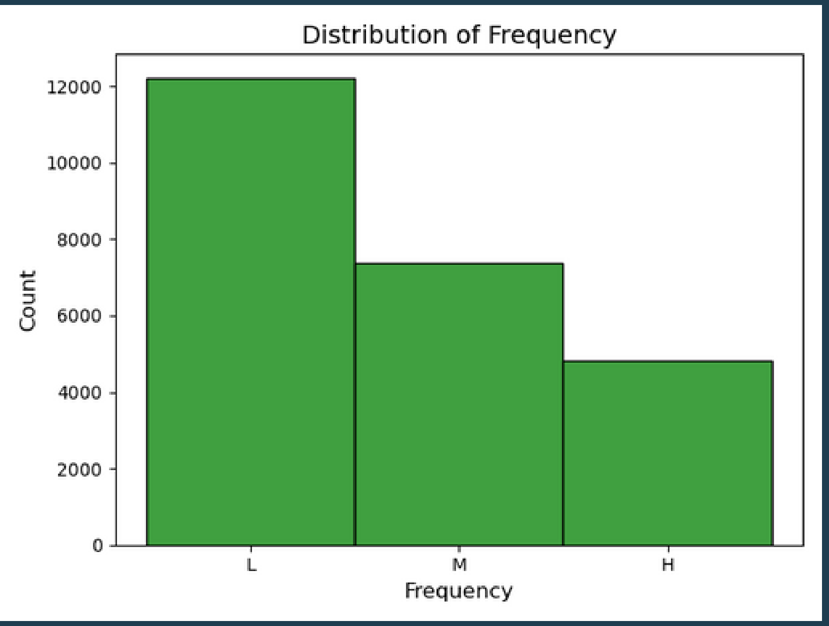
RFM Cluster

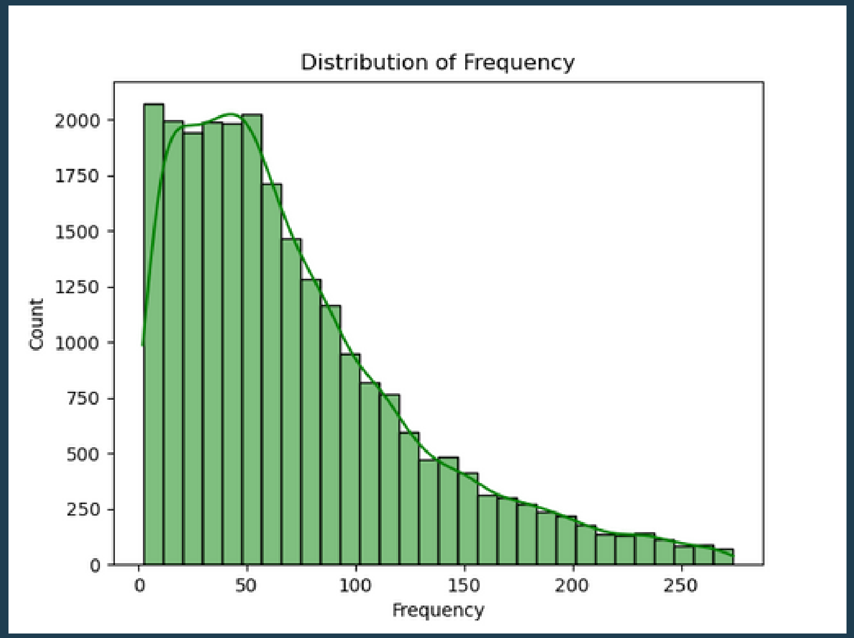
- **Champions**: High Frequency (F), high Recency (R), and high Monetary (M).
- **Loyal**: High Frequency (F), medium Recency (R), and medium Monetary (M).
- **Cannot Lose**: Medium Recency (R), medium-to-high Frequency (F), and medium-to-high Monetary (M).
- **Potential Loyalists**: High Recency (R), medium Frequency (F), and medium Monetary (M).
- **As New**: High Recency (R), low Frequency (F), and high Monetary (M).
- **To Reactivate**: Medium Recency (R), low Frequency (F), and medium Monetary (M).
- **Searching for Attention**: Medium Recency (R), medium Frequency (F), and low Monetary (M).
- **About to Sleep**: Low Recency (R), low-to-medium Frequency (F), and low-to-medium Monetary (M).
- **Bad**: Low Recency (R), low Frequency (F), and low Monetary (M).

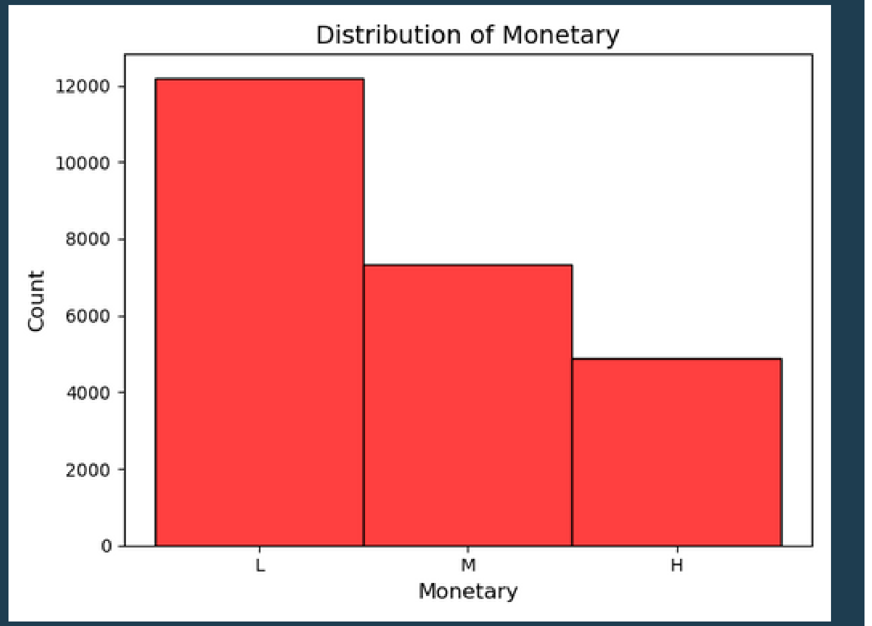
2.1 RFM ANALYSIS

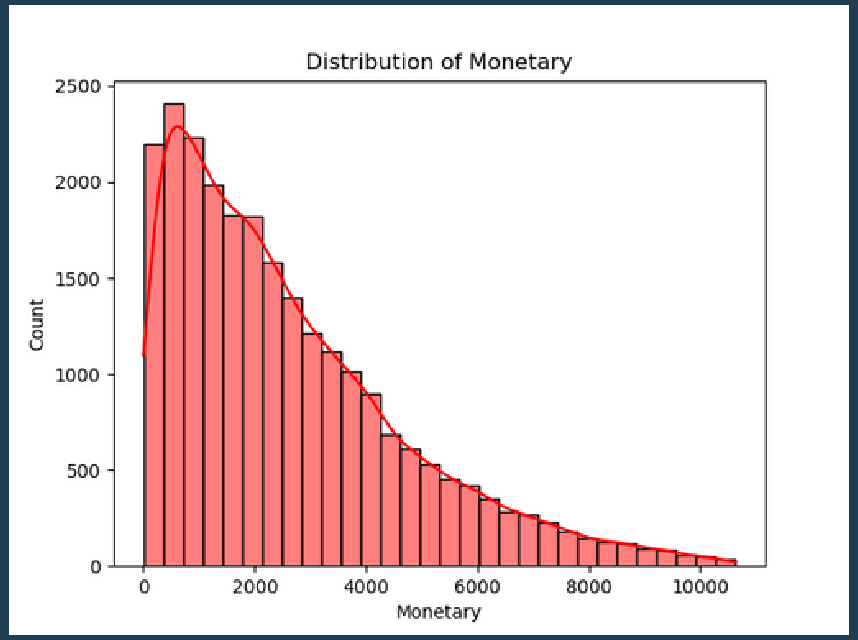












RFM Observation

This section analyses the distribution and meaning related to the basic RFM variables results obtained by the RFM analysis

RECENCY (0-5 Days: High, 5-12 Days Medium, 12+ Days Low)

- **Segmentation:** The **majority of the customers** fall into the **high recency** category, meaning they made recent purchases and are engaged with the business. However, **medium and low recency** groups, who are significantly **smaller**, might be the target of some marketing campaigns, aimed at re-engaging them.
- **Graph:** The continuous histogram shows an **exponential decay pattern** that justify the segmention results. The sharp decline as recency increases suggests the need for retention strategies to keep customers consistently engaged over time.

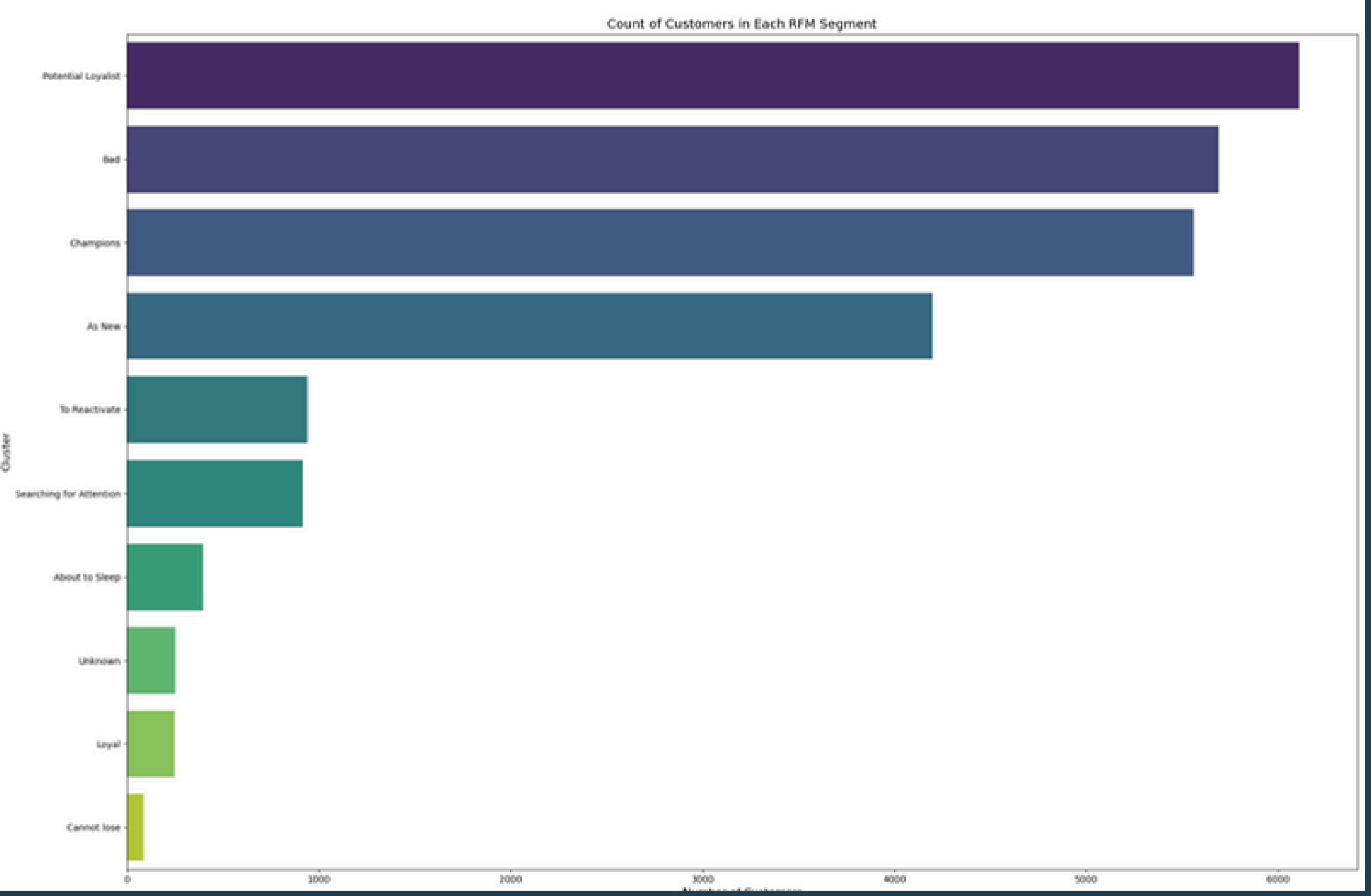
FREQUENCY

- **Segmentation:** There is a **substantial proportion** of customers in the **low category**. These customers have only made a few purchases and might represent new or less engaged buyers. The **medium category** shows a moderate count, while the **high category** represents loyal customers who make frequent purchases.
- **Graph:** The continuous frequency distribution exhibits a **long tail**, with most customers having a low and medium purchase frequency, while only a few exhibit a high frequency. This skew highlights the importance of converting low-frequency buyers into more frequent shoppers through loyalty programs or incentives.

MONETARY

- **Segmentation: Most customers** fall into the **Low spending** category, meaning they make small-value purchases, which could be attributed to entry-level or trial products. The **High spenders** are relatively **few** but are critical for profitability, representing a small but valuable segment to retain and grow.
- **Graph:** The continuous monetary distribution shows a **long-tail** pattern with most customers contributing lower revenue values. However, there is a significant minority of high-value customers who contribute disproportionately to the total revenue.This highlights the importance of personalized campaigns to nurture high-value customers and convert low-value customers into more profitable ones.

2.2 RFM SEGMENTS



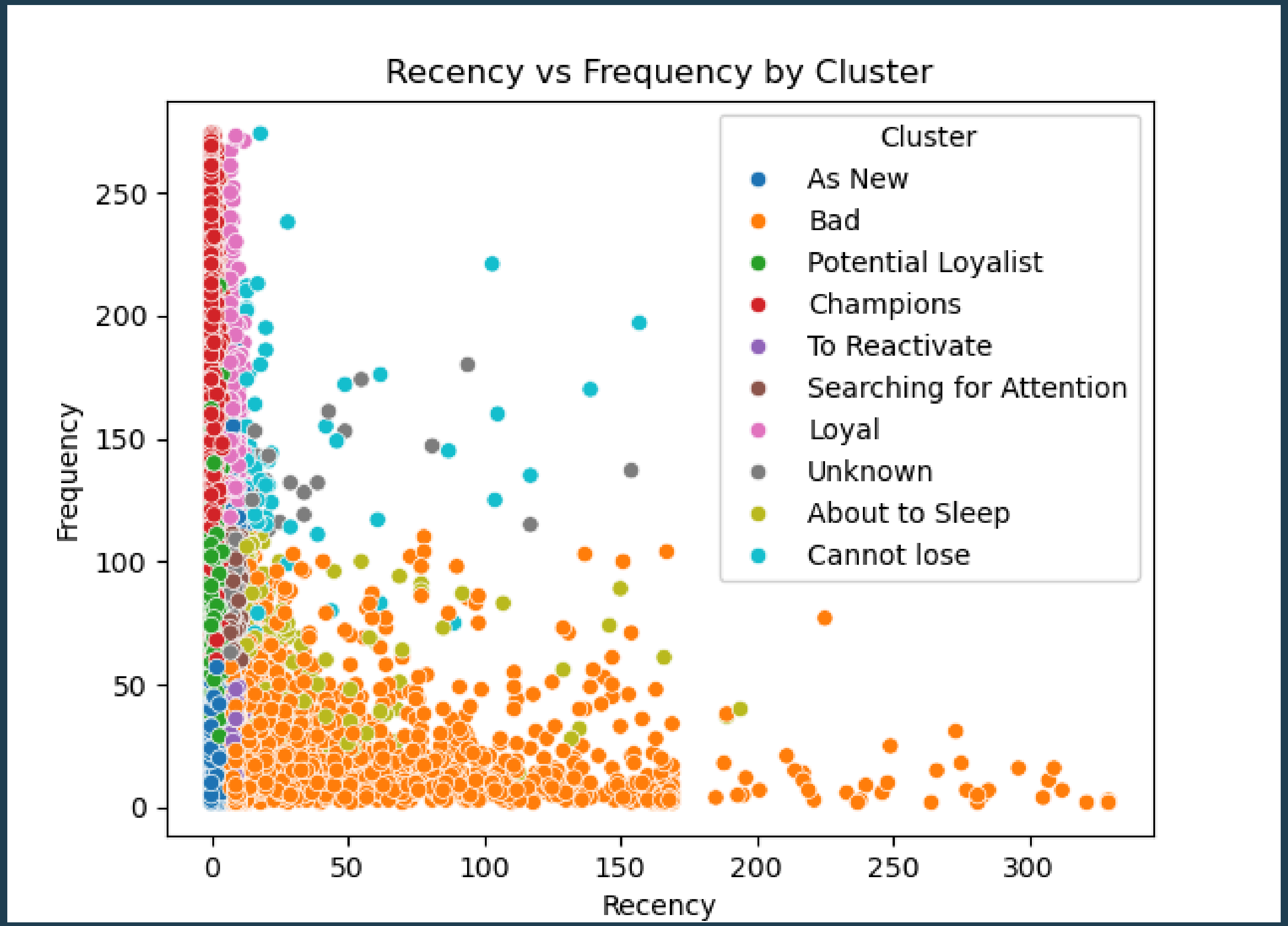
The chart shows the distribution of customers across various RFM (Recency, Frequency, and Monetary) segments. The RFM segments were derived based on customer behaviors and purchasing patterns. The primary insights from the chart are as follows:

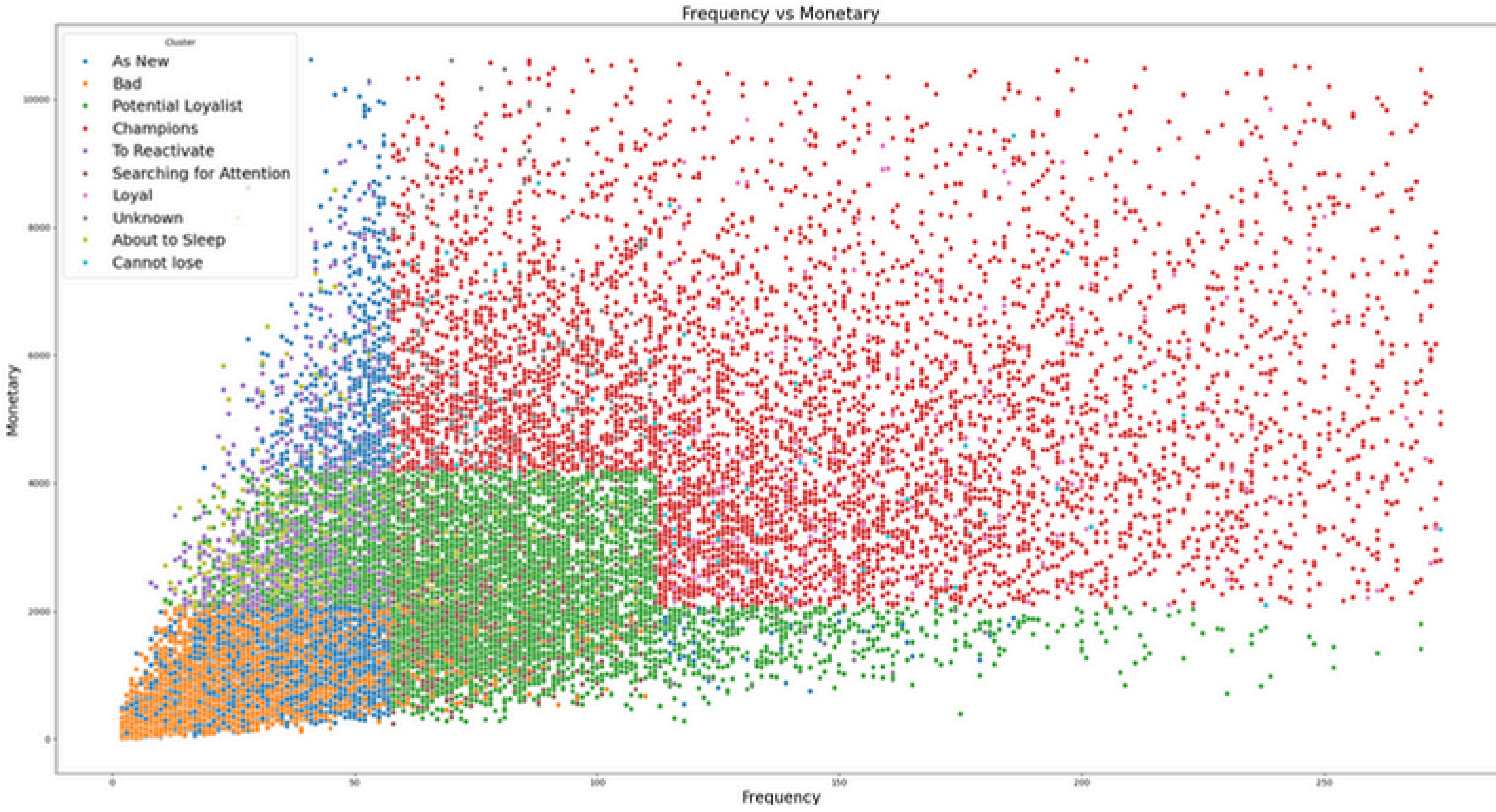
- "Potential Loyalist" is the largest segment, with over 6,000 customers. This group represents customers who are on the verge of becoming highly loyal based on their frequency and monetary contributions.
- The "Champions" segment is the second-largest, comprising loyal customers who frequently purchase and contribute significantly to the company's revenue.
- "Bad" and "As New" are other notable segments, indicating customers who might be inactive (Bad) or new entrants (As New).
- Smaller segments such as "Reactivate" and "Cannot Lose" suggest opportunities for targeted campaigns to either re-engage customers or retain high-value customers who are at risk of churn.
- The segments "About to Sleep" and "Unknown" have fewer customers, but they represent specific behaviors that might require unique strategies, such as understanding unknown patterns or preempting inactivity.

Strategic Recommendations

1. **Focus on "Potential Loyalist"**:
   - Nurture this group by offering loyalty programs, personalized recommendations, or exclusive deals to convert them into "Champions."
   - Monitor their behavior to identify trends or triggers that influence their purchasing decisions
2. **Re-engage "Bad" and "Reactivate" segments**:
   - Utilize targeted email or social media campaigns to reignite interest in your products.
   - Offer discounts or promotions to encourage purchases
3. **Appreciate "Champions"**:
   - Maintain strong relationships with "Champions" by recognizing their loyalty through rewards and exclusive perks
   - Gather feedback from this segment to enhance overall customer experience
4. **Monitor "About to Sleep"**:
   - Implement proactive strategies to re-engage these customers before they churn. Reminders, limited-time offers, or surveys might be effective

2.3 RFM CLUSTERING

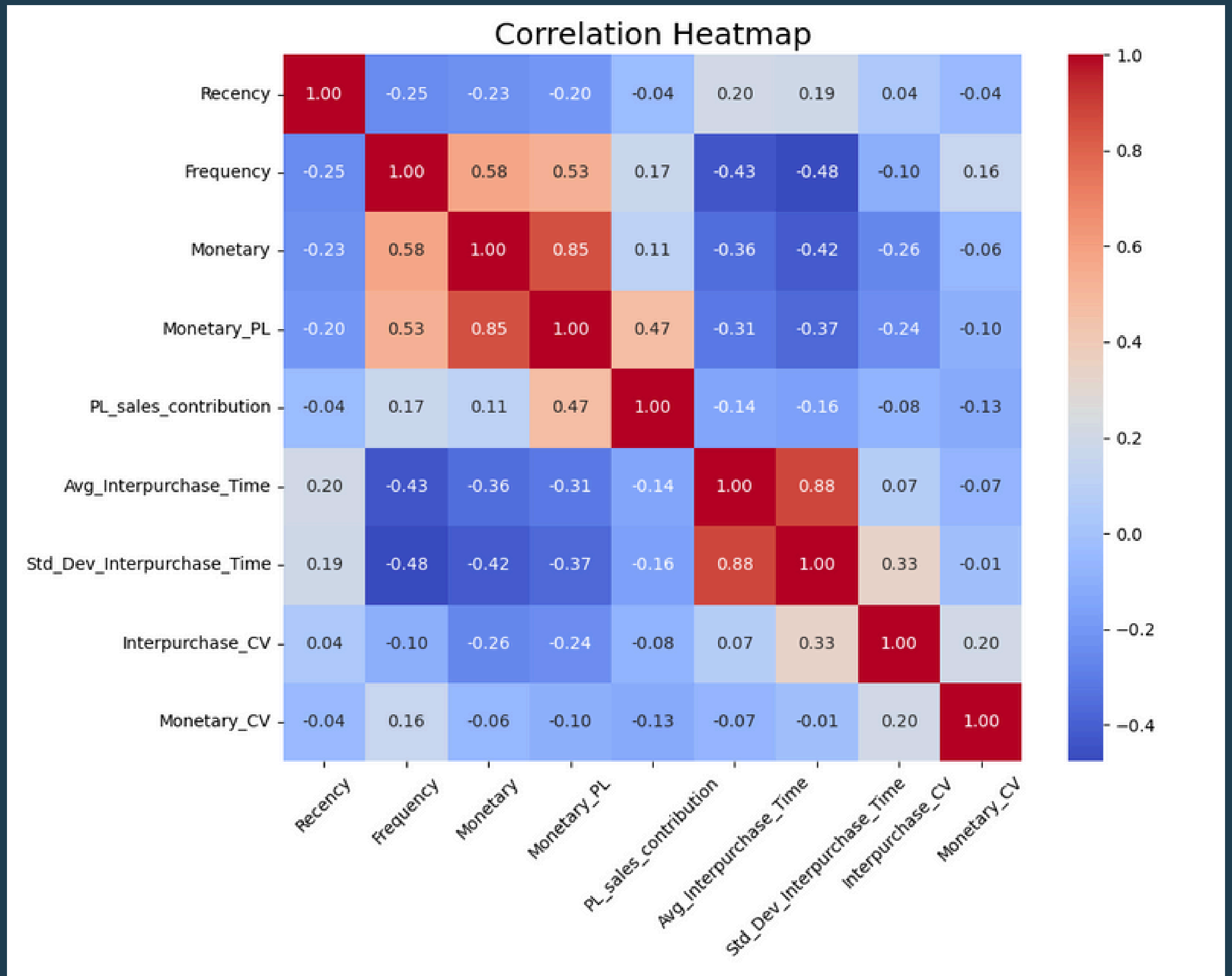




Customers in the "Champions" and "Potential Loyalists" clusters are densely populated near the lower recency and higher frequency region, indicating frequent and recent engagement with the brand. On the other hand, the "Bad" and "About to Sleep" clusters are concentrated in areas with high recency and low frequency, reflecting disengaged or at-risk customers. The "As New" cluster, as expected, shows lower frequency, often accompanied by low recency, representing newer customers who may have just started engaging with the brand. "Searching for Attention" and "To Reactivate" clusters are dispersed across mid-range recency and frequency values, reflecting inconsistent engagement. It is concluded that there is no strong correlation between rececy and frequency.

The scatter plot of Frequency vs. Monetary reveals key insights about customer behavior across clusters. Champions, represented in red, dominate the high-frequency and high-monetary areas, indicating their crucial role as top spenders and frequent buyers. Potential Loyalists, shown in green, occupy the mid-to-high range, presenting a strong opportunity to convert them into Champions through targeted offers. As New and Bad clusters, depicted in blue and orange, respectively, are concentrated in the lower frequency and monetary range, highlighting the need for onboarding efforts and re-engagement strategies. Loyal and Cannot Lose clusters, marked in purple and yellow, display moderate frequency and monetary values, making them ideal candidates for upselling and cross-selling. Meanwhile, clusters like To Reactivate and Searching for Attention, represented in pink and gray, show lower engagement levels, requiring reactivation campaigns to increase their activity

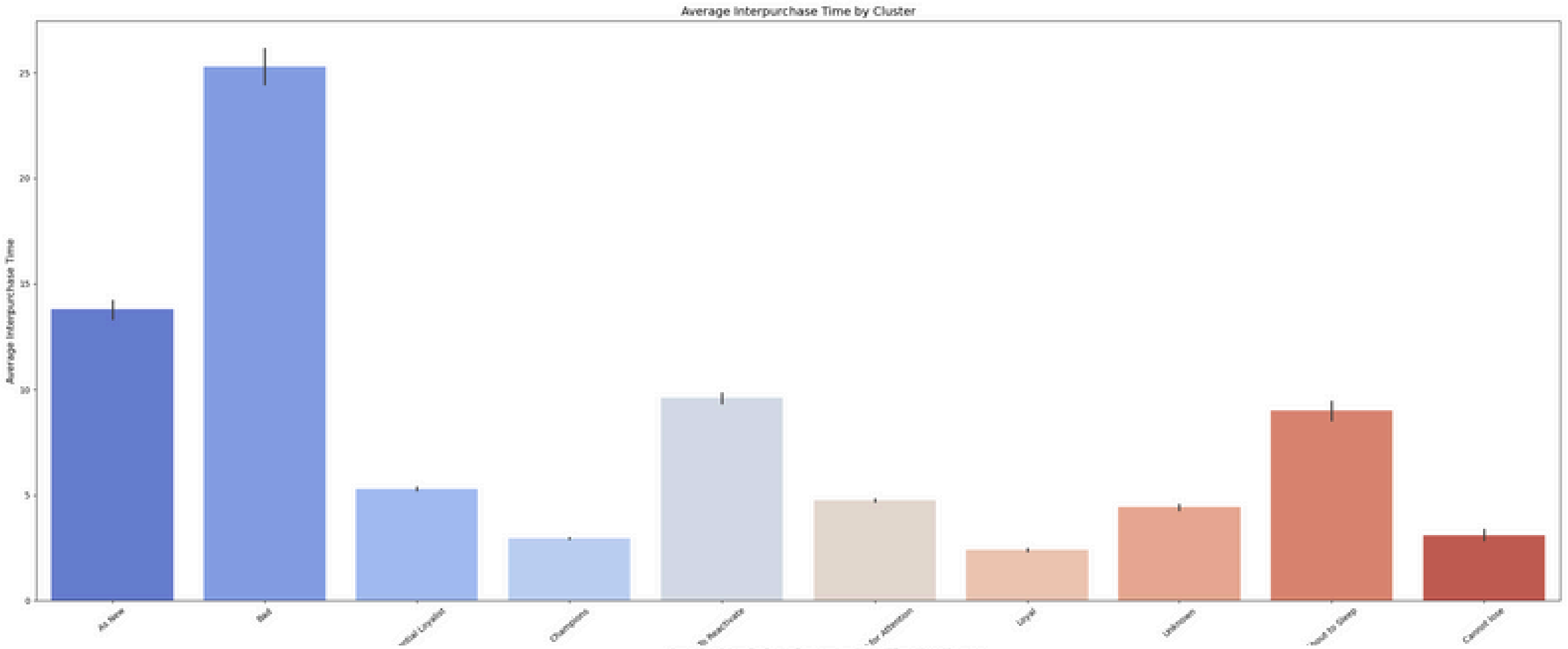
2.4 RFM ADDITIONAL ANALYSIS

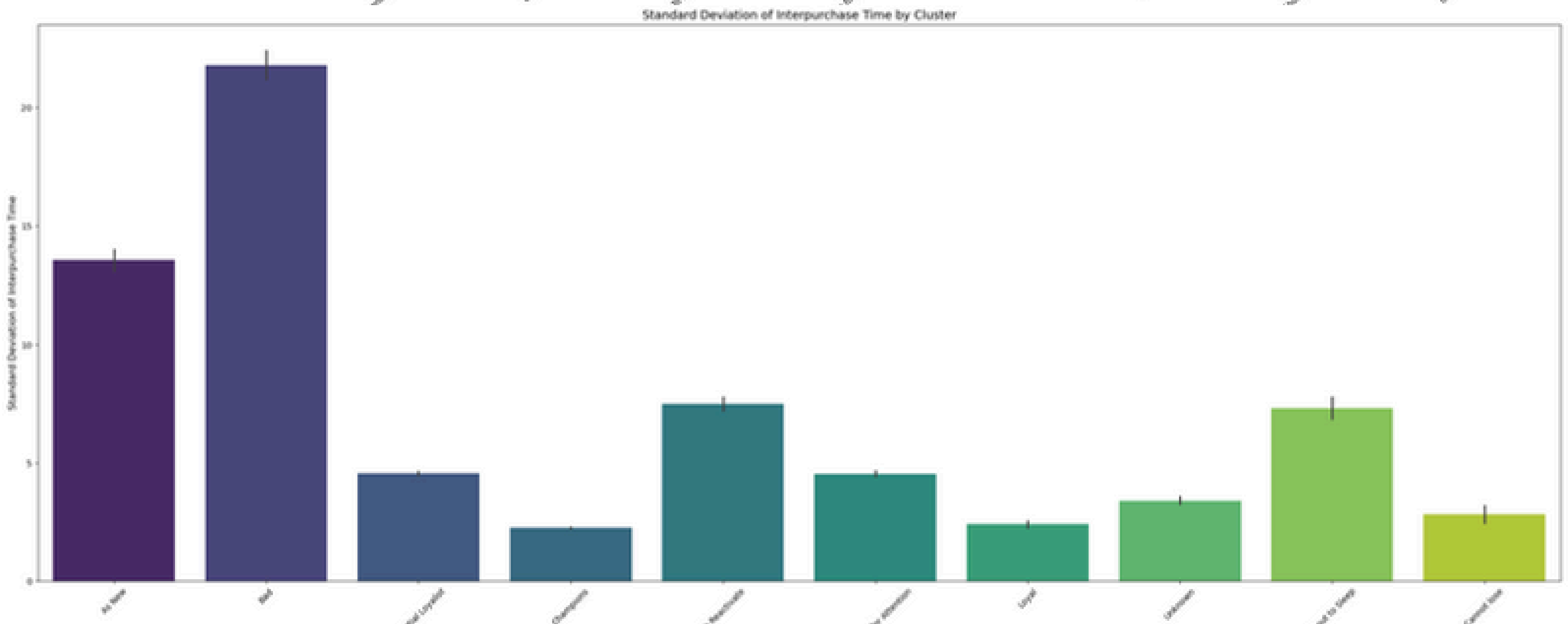


Frequency has a substantial positive connection with Monetary (0.58) and Monetary_PL (0.53), showing that regular purchasers contribute considerably to overall and PL sales. Monetary and Monetary_PL also have a strong association (0.85), highlighting the relevance of PL items in driving overall income.

The average interpurchase time exhibits a moderate negative connection with frequency (-0.43) and monetary (-0.36), indicating that shorter gaps between purchases are related with higher spending and purchase frequency. Similarly, the Standard Deviation of Interpurchase Time has a stronger negative association with Frequency (-0.48) and Monetary (-0.42), indicating that consistent purchasing behavior is associated with higher frequency and monetary value.

PL_sales_contribution has a moderate positive association with Monetary_PL (0.47), indicating its impact on overall sales.





Clusters such as "As New" and "Bad" have the highest average interpurchase time, indicating infrequent purchases and a need for re-engagement strategies. The standard deviation is also high for these clusters, reflecting inconsistent purchasing behavior.
"Champions," "Potential Loyalists," and "Cannot Lose" have the lowest interpurchase times, signifying frequent and reliable purchasing habits, indicative of their loyalty and value to the business.

Clusters like "To Reactivate" and "Searching for Attention" exhibit moderate interpurchase times and variability, signaling opportunities for tailored campaigns to increase purchase frequency. Clusters with low variability, such as "Cannot Lose" and "Champions," indicate steady behavior, while higher variability in "As New" and "Bad" suggests erratic patterns that require more dynamic engagement strategies.

4. MBA - METHODOLOGY

The MBA was done by iterating over Support and Confidence values to identify a combination that balances predictive power (Confidence ≥ 70%) and manageable rule sets (10-20 rules). At each cycle we excluded high-centrality categories (present in ≥ 5 rules) to uncover less frequent but still meaningful patterns and reduce potential spurious correlations. The whole cycle was repeated 4 times.

Identify the best values

We identified the best combination of Support and Confidence to balance predictive power and a manageable rule set.

Interpret the results

We interpreted the rules to derive insights about purchasing patterns and draw recommendations.

Remove Categories with high centrality

We removed Categories with high centrality to uncover less frequent but still meaningful patterns.

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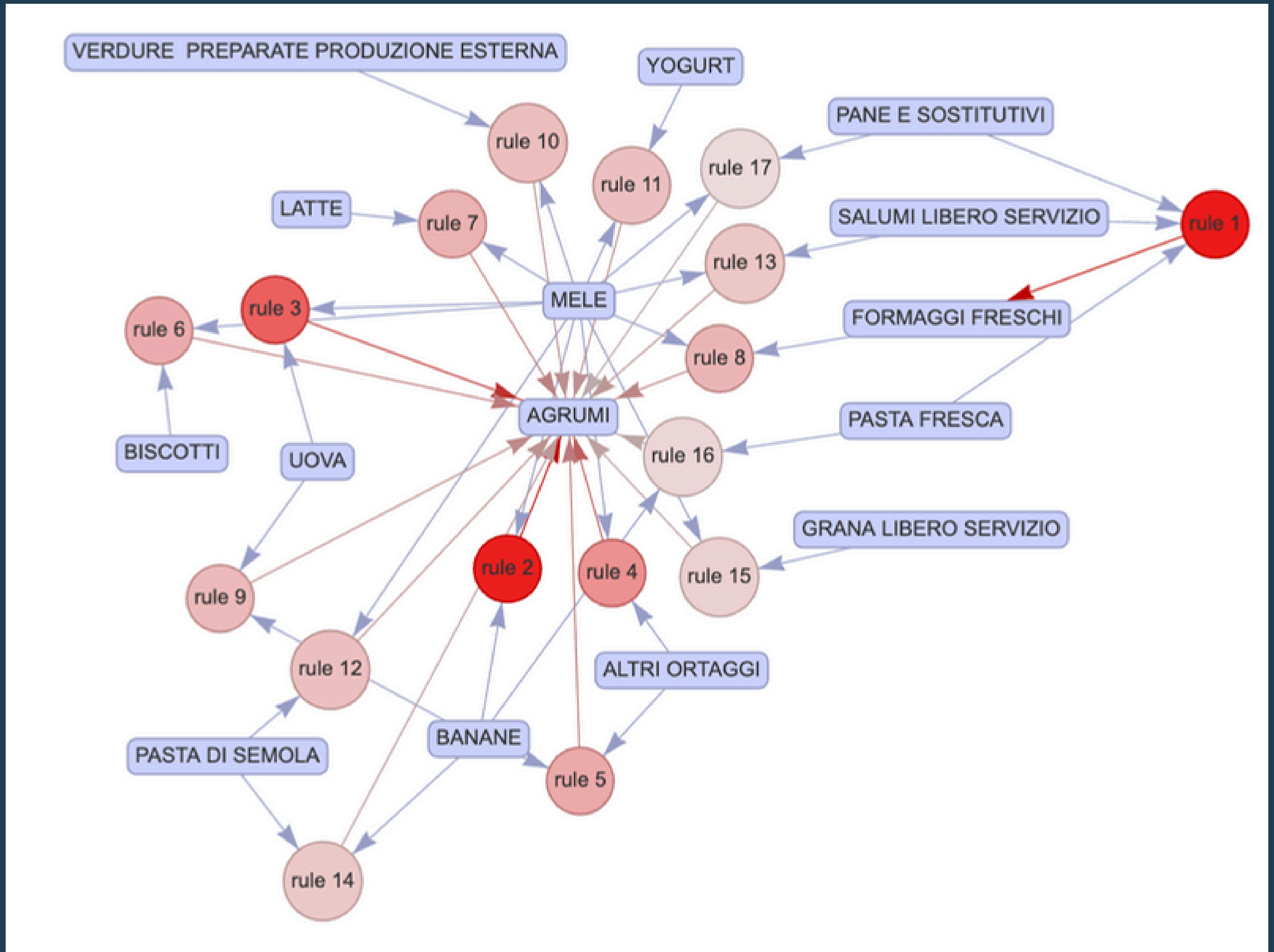
4.1 MBA - FIRST CYCLE



Most of the rules are mediated by either **AGRUMI** or **MELE**, which are both very common in customer purchases. **BANANE** also has a high centrality, suggesting that Categories in the **Fruit Section** are common purchases and are often bought with other items.

SUPPORT: 2,5%
CONFIDENCE: 70%
RULES FOUND: 17

Interesting Rules

1) PANE + SALUMI + PASTA FRESCA => FORMAGGI
A consistent percentage of customers either does not cook or has very little time to cook.

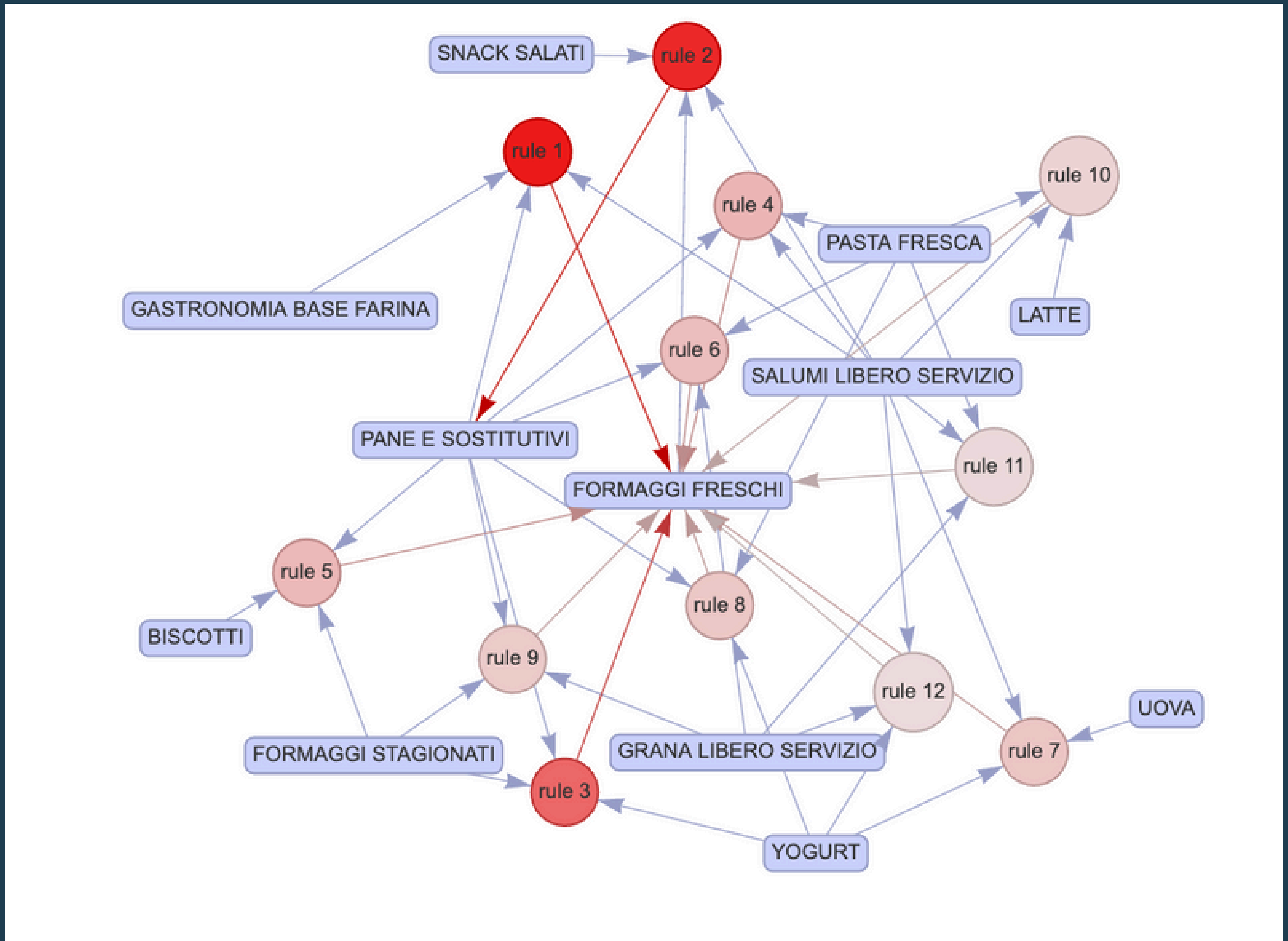
2) BANANE + MELE => AGRUMI
A consistent percentage of customers consumes lots of different fruits and is seeking variety within the category.

3) BANANE + ORTAGGI => AGRUMI
People who consume a high amount of fruit also consume a high amount of vegetables.

Categories Removed

- AGRUMI
- MELE
- BANANE

4.2 MBA - SECOND CYCLE



Here most of the high-centrality Categories can be consumed as a **Quick Meal**, like **PANE**, **FORMAGGI FRESCHI**, **FORMAGGI STAGIONATI**, **GRANA LIBERO SERVIZIO**, **SALUMI LIBERO SERVIZIO**, **PASTA FRESCA**.

SUPPORT: 2,0%
CONFIDENCE: 70%
RULES FOUND: 12

Interesting Rules

1) FORMAGGI FRESCHI + SALUMI + SNACK => PANE
Could be the consequence of a fast (and unhealthy) lifestyle with no-cooking required.

2) GASTRONOMIA + SALUMI + PANE => FORMAGGI
Consistent use of pre-cooked food and quick meals.

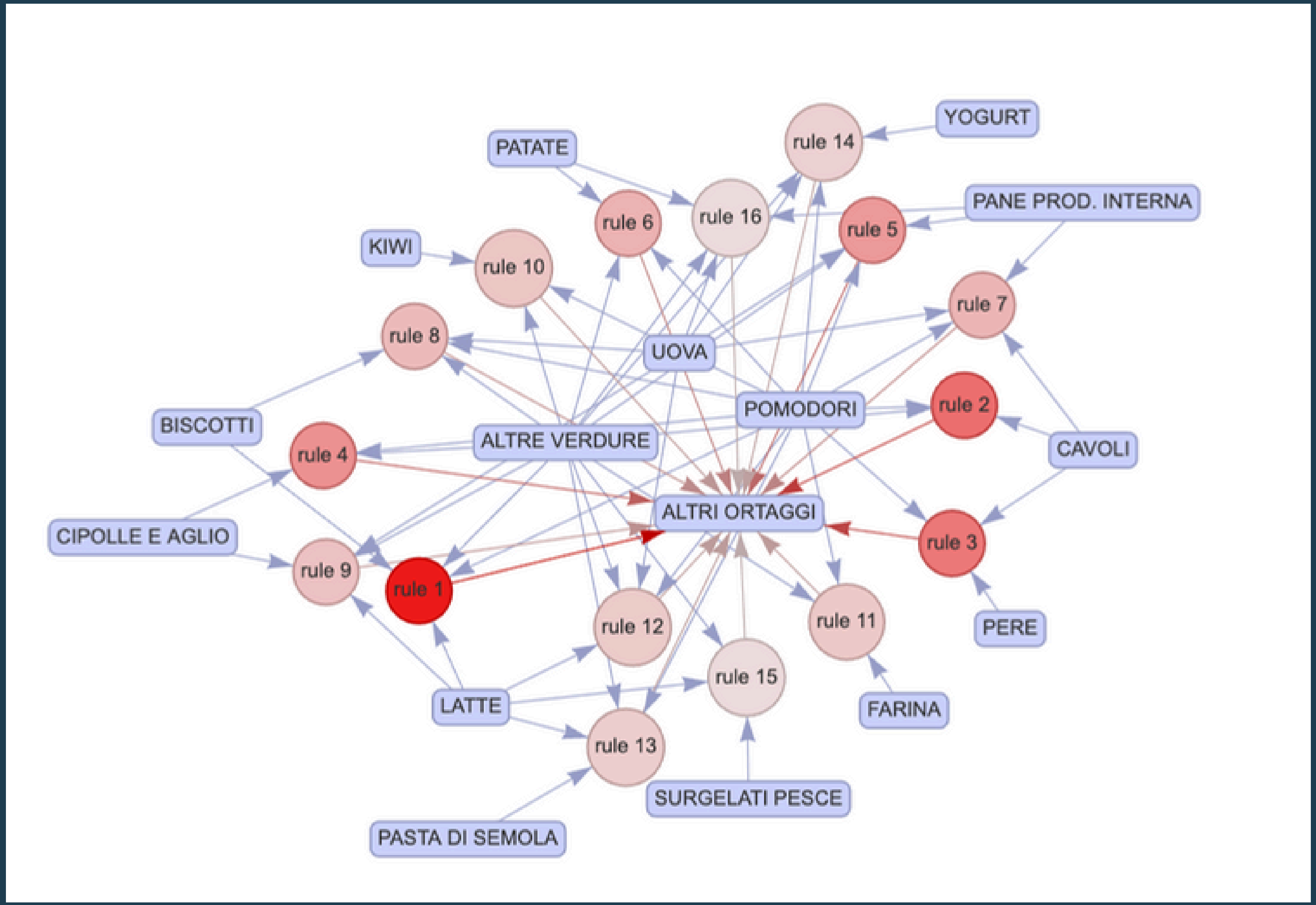
3) F.STAGIONATI + PANE + YOGURT => F.FRESCHI
Large consumption of diary products.

4) GRANA + PANE + PASTA FRESCA => F.FRESCHI
Mediterranean diet with large use of pasta with cheese.

Categories Removed

- FORMAGGI FRESCHI
- SALUMI LIBERO SERVIZIO
- PASTA FRESCA
- PANE E SOSTITUTIVI

4.3 MBA - THIRD CYCLE



This third iteration shows that the most popular items come from the **Vegetable Section**, like **ALTRI ORTAGGI**, **ALTRE VERDURE**, **POMODORI** and **Animal Derivatives** like **LATTE** and **UOVA**.

SUPPORT: 0,275%
CONFIDENCE: 77,5%
RULES FOUND: 16

Interesting Rules

1) VERDURE + BISCOTTI + LATTE + POMODORI => ORTAGGI
Breakfast at home and at least one meal based on varied vegetables.

2) VERDURE + CAVOLI + POMODORI => ORTAGGI
Diet with consistent use of many different vegetables.

3) VERDURE + PERE + POMODORI => ORTAGGI
Correlation between high consumption of fruit and vegetables.

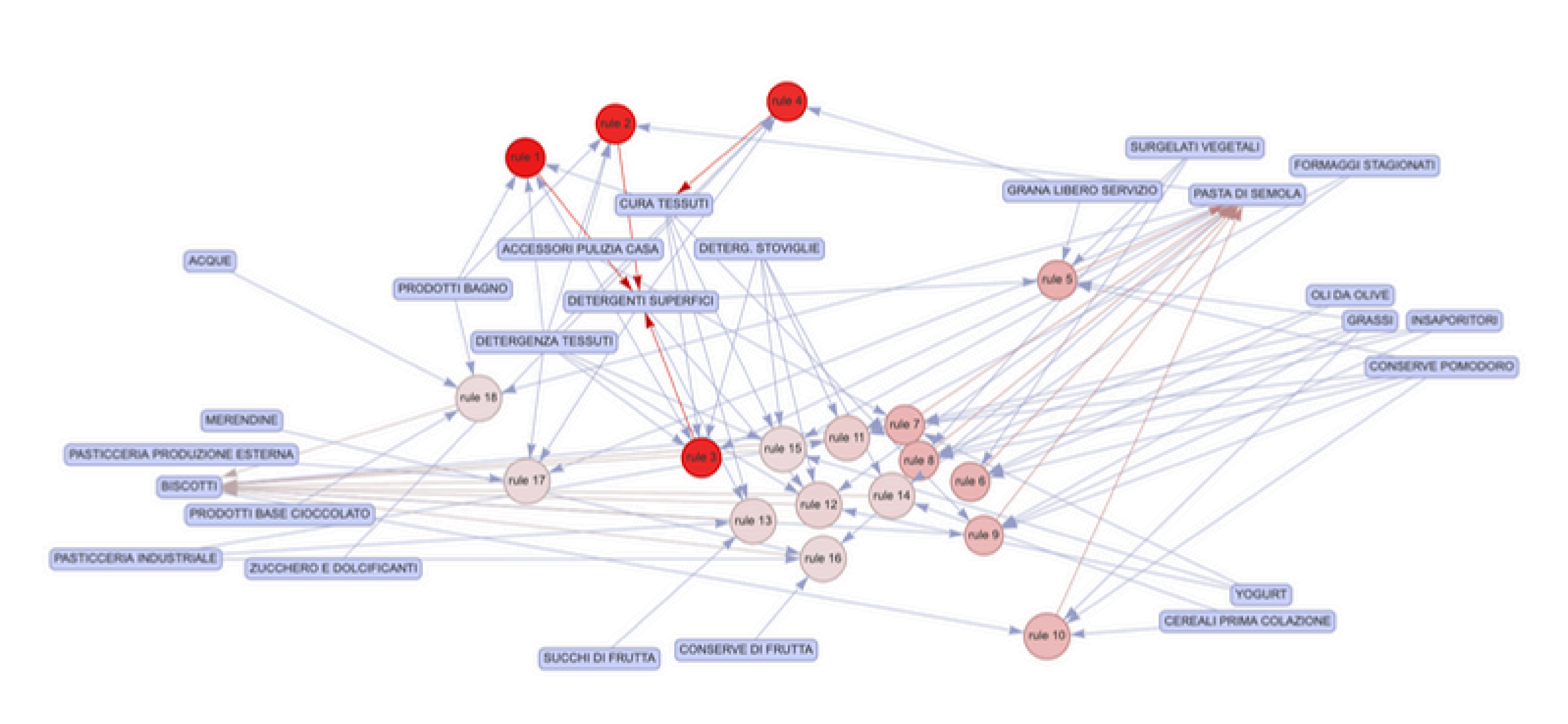
4) VERDURE + CIPOLLE + LATTE + UOVA => ORTAGGI
Vegetarian diet with non-meat proteins.

Categories Removed

- POMODORI
- ALTRI ORTAGGI
- ALTRE VERDURE
- UOVA
- LATTE

4.4 MBA - FOURTH CYCLE

SUPPORT: 0,2125%
CONFIDENCE: 73%
RULES FOUND: 18



This final iteration shows many independent clusters: **Sweets**, **Fruit Products**, **Cleaning Products**, **Breakfast Food** and **Condiments**.

Interesting Rules

1) PULIZIA CASA + CURA TESSUTI + DETERGENZA TESSUTI + PRODOTTI BAGNO => DETERGENTI SUPERFICI
Many cleaning products, although for different scopes, are often bought together.

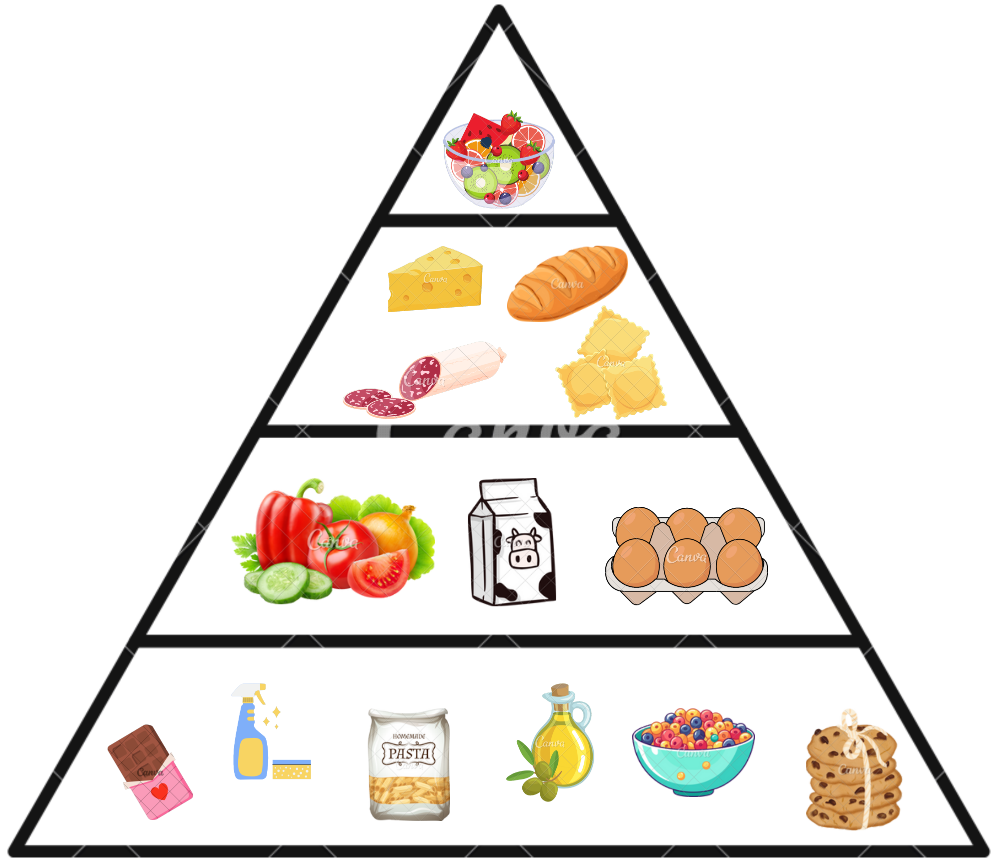
2) CONSERVE POMODORO+ BISCOTTI + CEREALI + GRASSI => PASTA DI SEMOLA
Diet based on wheat products with fast cooking and long conservation.

3) CONSERVE FRUTTA + MERENDINE + PASTA DI SEMOLA + PASTICCERIA INDUSTRIALE => BISCOTTI
Diet based on processed and packaged carbohydrates.

Categories Removed

- PASTA DI SEMOLA
- DETERGENTI SUPERFICI
- DETERGENZA TESSUTI
- CURA TESSUTI
- DETERGENTI STOVIGLIE
- BISCOTTI
- GRASSI
- CONSERVE DI POMODORO

4.5 MBA - RECOMMENDATIONS



GENERATORS OF RULES

BANANE - MELE - SALUMI - PASTA FRESCA - PANE - POMODORI - LATTE - UOVA - VERDURE - CURA TESSSUTI - DETERGENZA TESSUTI - DETERGENTI SUPERFICI - DETERGENTI STOVIGLIE - GRASSI - CONSERVE DI POMODORO

CONSEQUENCE OF RULES

AGRUMI - FORMAGGI FRESCHI - ORTAGGI - PASTA DI SEMOLA - BISCOTTI

MARKETING STRATEGIES

Product Hierarchy

- We can map a hierarchy of the products purchased by our customers thanks to the categories which have been removed at each cycle.

Product Placement

- Group Quick Meal Items (Pane, Salumi, Formaggi Freschi, Pasta Fresca) in one area for convenience.
- Arrange Fruits and Vegetables (Banane, Mele, Agrumi, Ortaggi, Verdure) to address health-sensitive customers .
- Create a dedicated section for Cleaning Products (Detergenti Superfici, Cura Tessuti, Detergenti Stoviglie) to simplify bulk buying.

Bundling

- Offer Pre-Prepared Sandwich Kits combining Pane, Formaggi, and Salumi.
- Provide Cleaning Essentials Packs with Detergenti, Cura Tessuti, and Prodotti Bagno.

Promotions

- Highlight Pre-Cooked Foods (Pane, Salumi, Pasta Fresca) for time-strapped customers.
- Set up Tasting Stations for complementary products like Formaggi Freschi and Salumi.

Inventory Management

- Prioritize stock for categories which are generator of rules.
- Optimize availability for categories which are consequence of rules.

5. CHURN ANALYSIS

DATASET PREPARATION

The goal of this analysis is to be able to predict customer churn by understanding the factors influencing it. The process consisted in trying three different models to identify the key variables for churn prediction, choosing the one with best performances. The identified variables were thene analysed with relation to time. To do so, we started with transactional data from the original dataset (Type B) with RFM segmentation categories and applied pecific data preparation techniques.

1. COLUMNS ADJUSTMENT

Transactions were **grouped by id_customer and date** to compute aggregated metrics of existing columns or to create new ones belonging to the following categories:

- **General**: number of orders (total_orders), unique stores visited (different_stores), most frequent store type (main_store_type) and number of days since the customer's first recorder purchase (tenure_days)
- **Sales**: total private label gross sales (PL_gross_sales), total gross sales and average gross sales
- **Items**: total and average number of private label items (total_item_PL, avg_items_PL) and non-private label items (total_items_other, avg_items_other).

Subsequently, three columns were added from the **RFM Analysis' results dataset**, namely:

- **Avg_Interpurchase_Time**: average number of days between two purchases
- **R**: classification column for Recency (values: L, M, H)
- **F***: classification column for Frequency (values: L, M, H)

** F was not added from the beginning, but later on for the survival analysis*

SIDE NOTE

General metrics were created to analyze customer behavior and loyalty, enriching customer profiles to increase probabilities of retrieving meaningful insight from the analysis

2. CHURN LABEL

Being a non contractual setting, the original dataset was missing any information abouth whether a customer had churned or not. Thus, we leveraged on the results of our RFM Analysis to build a binary churn variable.

- Customers with **low recency** (R = "L") were labeled as **churners** (churn = 1)
- Customers with **medium** (R = "M") **or high recency** (R = "H") were labeled as **non-churners** (churn = 0)

PROS OF THIS APPROACH

Recency is a good indicator of churn, as customers who have not purchased recently are less likely to be still active. Using RFM segmentation ensures the target variable is directly linked to past purchasing behavior, making it both relevant and actionable.

CONS OF THIS APPROACH

Being the churn variable artificial, a probability would have been more realistic than a binary variable. However, due to insufficient data, the computation of the probability gave results very similar to those of the binary variable so the first method was used, because more clear and direct.

3. OTHER DATA PROCESSING

In addition to **NAs check and removal**, a **correlation check** was made to avoid biases in the following analysis. Variables with an absolute correlation value > 0.75 were removed, and in particular:

- **total_gross_sales** and **total_PL_sales**: because a similar information about value of the purchases could be found in average sales value, that could be more helpful in eventually detecting future churn behaviors
- **total_items_other** and **avg_items_other**: because they had high correlation with PL items variables, to which was given a higher priority in the choice of the variables to keep

Lastly, the categorical variable main_store_type was transformed into a **dummy variable**

5. CHURN ANALYSIS

METHODOLOGY

TRAIN-TEST SPLIT

Training set - 70%
To build and optimize the models

Test set - 30%
To evaluate models' performances

CLASS IMBALANCE

The inspection of the dataset pointed out a class imbalance problem, which is coherent with RFM analysis clusters (>80% of customers classified as non-churners). To obtain solid results, **undersampling** was used at the beginning of each analysis

MODELS USED

1

LOGISTIC REGRESSION

Reasons: well-suited for predicting binary outcomes; first exploratory attempt.

Results: Accuracy: 0.68 | Precision: 0.25 | Recall: 0.80

Conclusions: Very low precision. Attempt with another model

2

DECISION TREE

Reasons: easy to interpret and visualizes the decision-making process.

Results: Accuracy: 0.75 | Precision: 0.29 | Recall: 0.70

Conclusions: Better Accuracy and Precision but much lower Recall. Attempt with a more powerful model

3

XGBOSST

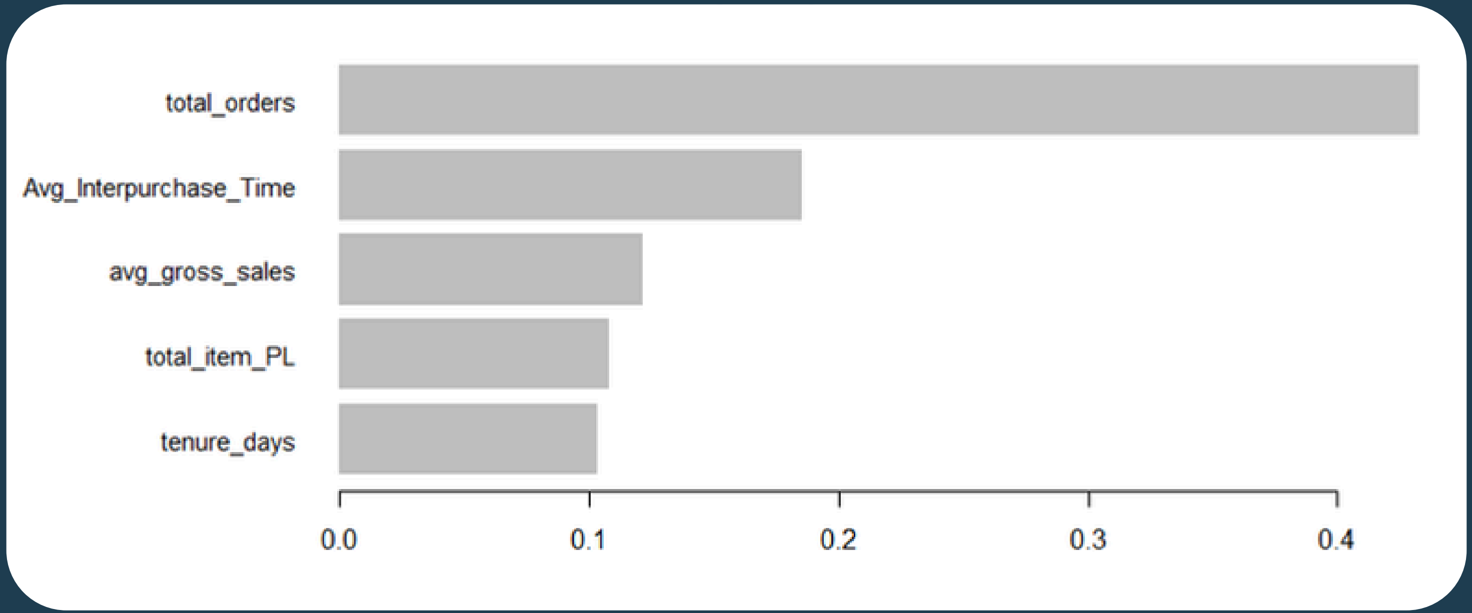
Reasons: powerful, gradient-boosting algorithm that handles large datasets efficiently and can capture more complex relationships.

Results: Accuracy: 0.76 | Precision: 0.31 | Recall: 0.78

Conclusions: Good compromise with the other two models' results

XGBOOST INSIGHTS

MOST IMPORTANT VARIABLES



According to XGBoost model, these are the **top 5 variables** that can help in predicting churn.
The results are overall coherent with those of the other models used. The difference lies in the entity of the effect they have (positive/negative)

EFFECT OF THE VARIABLES ON CHURN PROBABILITY

To evaluate whether the variables had a positive or negative effect on churn probability, we computed the **SHAP values** for each.
As the summary showcases, for the abovementioned variables we can say that:

- **Total Orders** and **Total Items PL** have overall a **negative** effect, meaning that they reduce probability of churn as they increase.
- **Avg Interpurchase Time** and **Tenure Days** have overall a **positive** effect, meaning the opposite as before
- **Avg Gross Sales**, is the most ambiguous one since there are two negative values but they are both lower than the positive one.

To better investigate this last point and to understand how churning probability varies considering both the variables and the customer "life" (i.e. tenure days), we conducted a survival analysis

```
> shap_summary
   total_orders different_stores ma
1:    -2.029764     -0.005167234
2:     9.666045      0.8319793
3:    -9.474817     -0.8835101
   total_item_PL avg_gross_sales av
1:    -0.2147277     -0.04566685 -0.
2:      1.744582       2.781271
3:     -2.952068      -1.877138
```

```
main_store_type tenure_days
    -0.02624384   0.2019642
      0.995675     1.99692
     -1.373185   -9.320755
avg_items_PL Avg_Interpurchase_Time
-0.004740919               0.863313
    1.248371                9.67684
    -1.02047              -5.591179
```

5. CHURN ANALYSIS

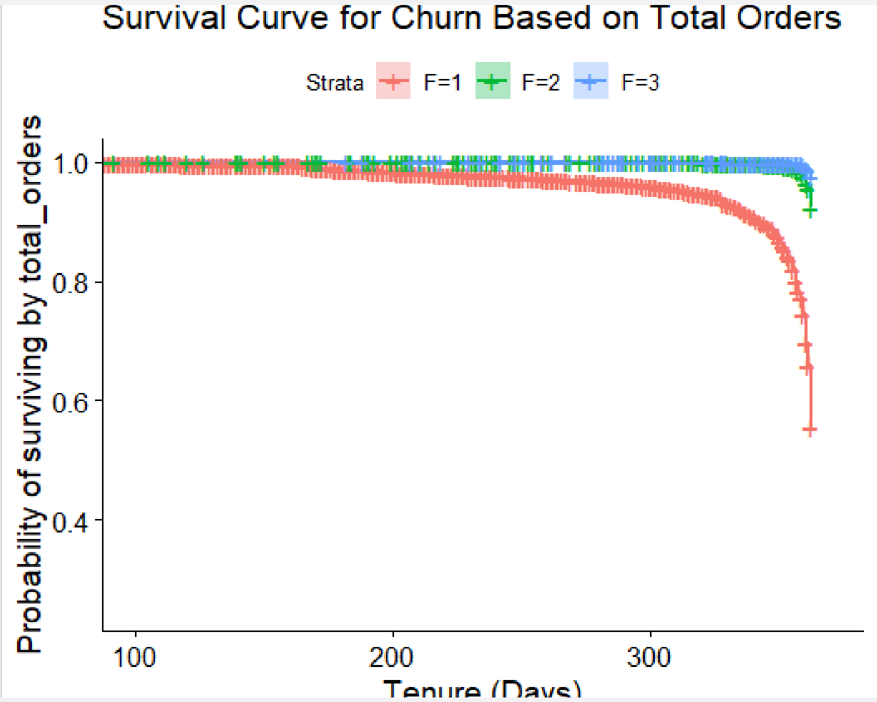
SURVIVAL ANALYSIS

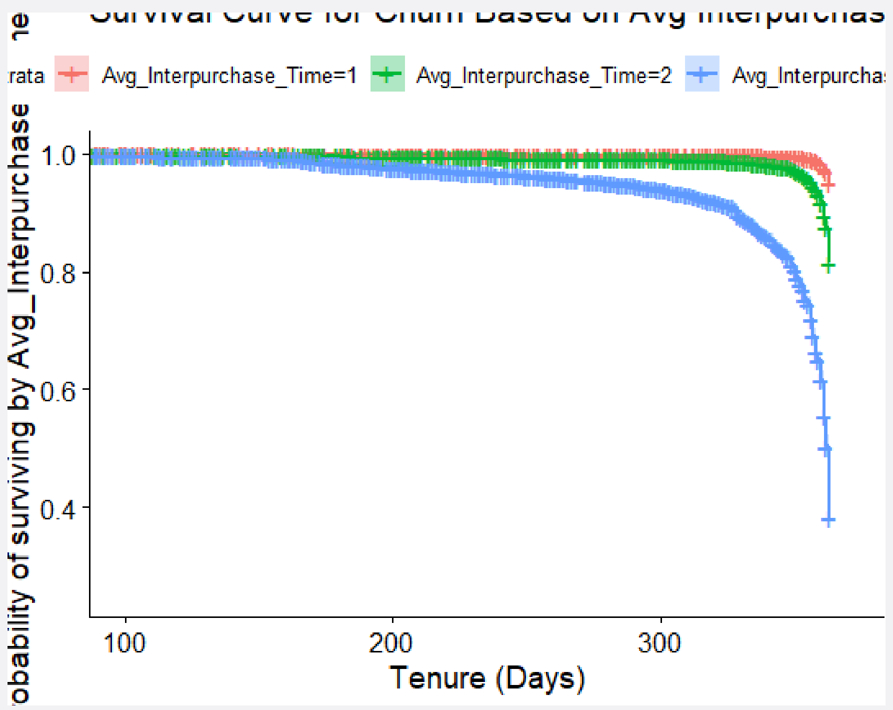
PRESETTING

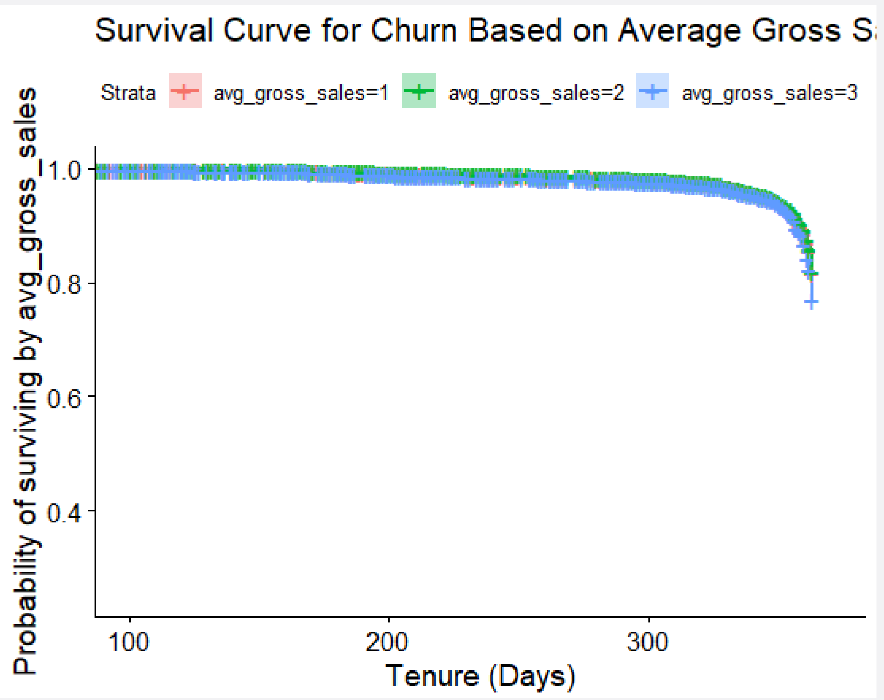
To conduct the analysis, three categories (1=Low, 2=Medium, 3=High) for each variable were created, according to the following criteria:

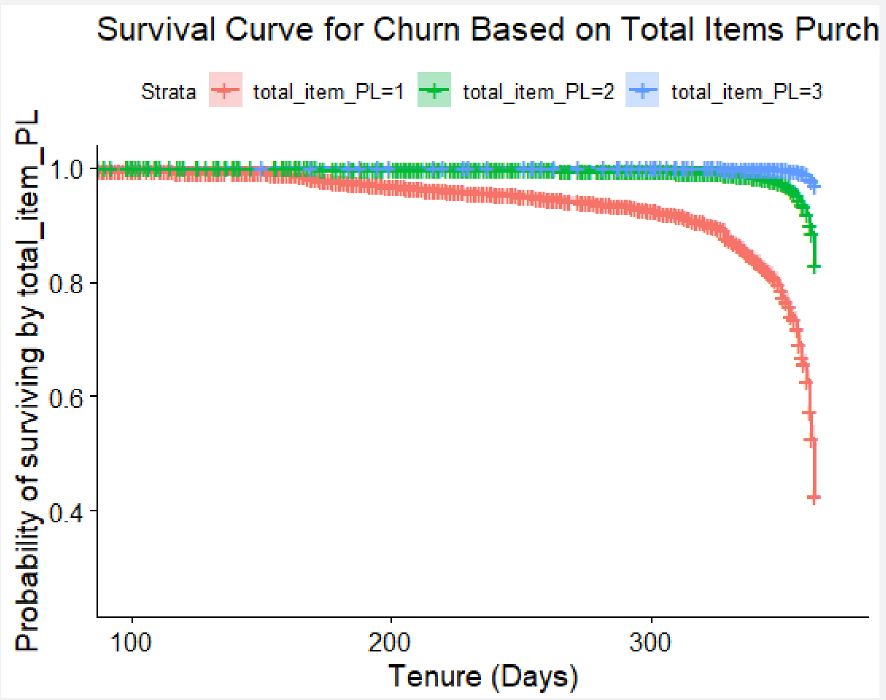
- **total_orders**: being the equivalent to frequency, we used the classification done in the RFM
- **Avg_Interpurchare_Time:** set manually: Low x<5 days | Medium 5<=x<=10 | High x>10
- **avg_gross_sales** and **total_item_PL**: based on distribution as in the code

RESULTS Total Orders | Avg Interpurchase Time | Avg Gross Sales | Total Items PL









TAKEAWAYS

1. **LESS ENGAGED PEOPLE CHURN EARLIER**: The less engaged people are for example those who didn't purchase many times **(low total_orders)** or those who wait much time between two consecutive purchases (**high avg_interpurchase_time**)

2. **AMBIGUOUS EFFECT OF GROSS SALES:** The survival analysis didn't clarify the doubt about whether **avg_gross_sales** has a positive or negative effect. However, it's reasonable to consider as reliable the results of the **Logistic Regression**, that suggests a slightly overall **negative effect**. Indeed, a person that spends more on each purchase, is more likely to be a loyal customer and thus less prone to churn.

3. **EFFECT OF PRIVATE LABELS:** As observed, the number of private label items purchased appears to have a negative effect on churn probability, indicating that they are a good predictor and element to leverage on in order to **reduce churn**. This insight opens the door to a range of marketing strategies, since the retailer has easier control and influence on these brands.

To offer more actionable insights and business recommendations, we have decided to perform a **deeper analysis of private labels**, focusing on the **differences between churners and non-churners**.

6. PRIVATE LABELS IN LOYALTY MECHANISMS: ANALYSIS AND INSIGHTS

Datasets used:

The analysis combines two datasets: data_churn_dummies (with churn labels) & rfm_analysis (with customer clusters)

Churn-Based PL analysis

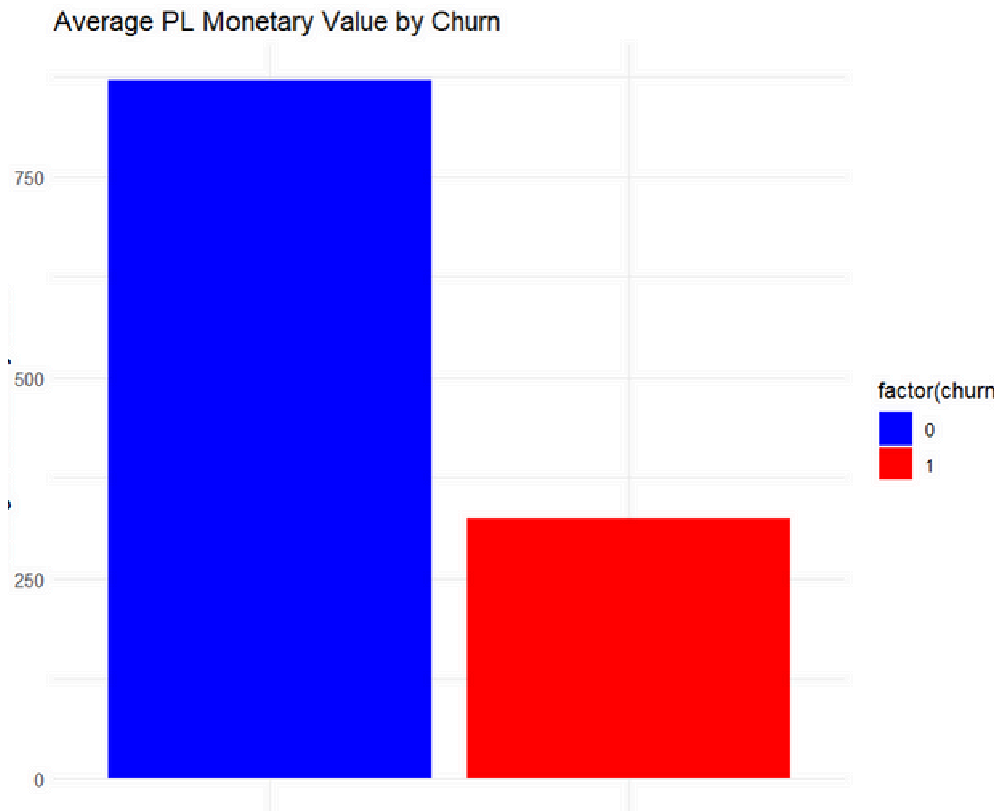
Studied average monetary value, item purchases, and PL contribution to total sales for churners (1) vs. non-churners (0) and we can see non-churners spend avg 870 on PLs compared to churners at 324, with higher PL contribution (31% vs. 28%)

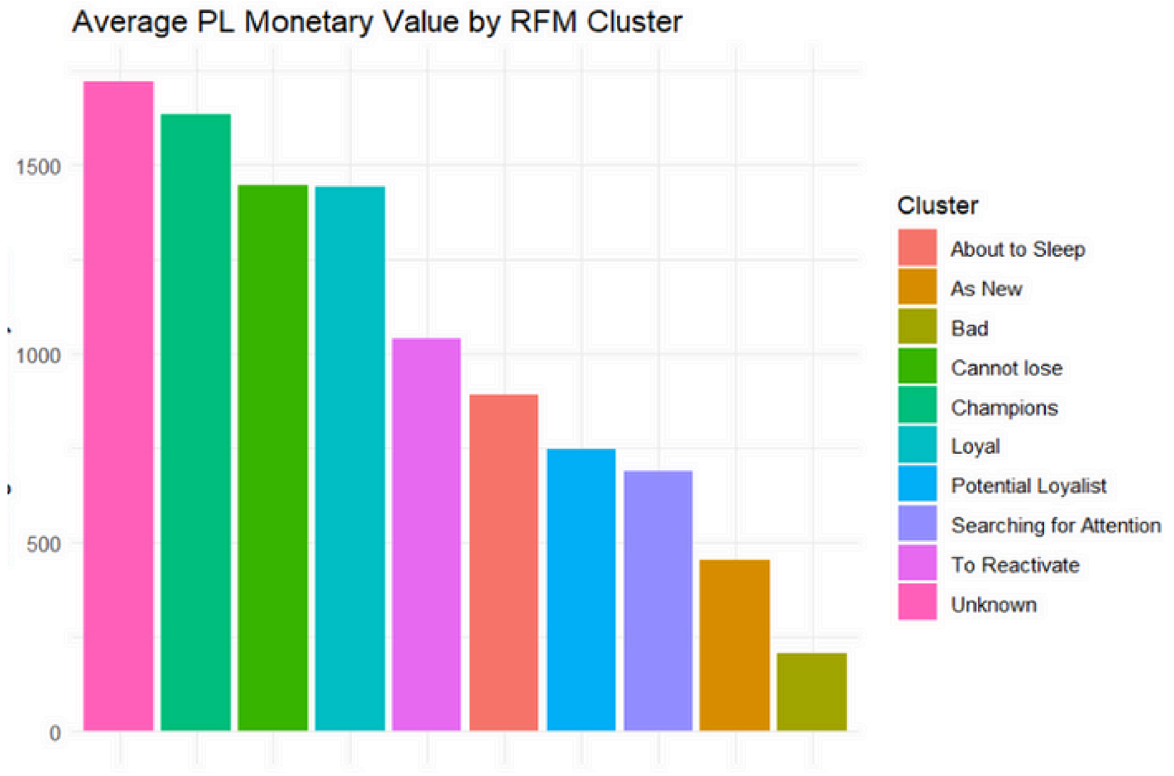
Cluster-Based PL analysis

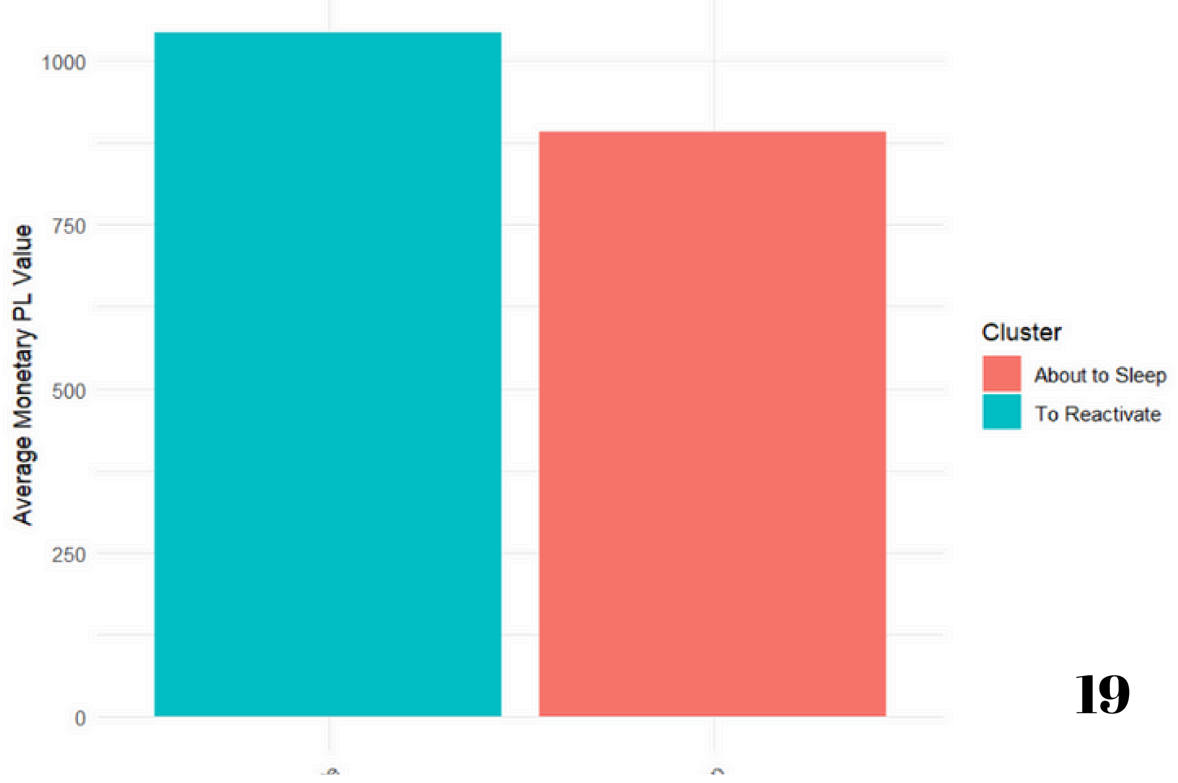
Compared PL engagement across RFM clusters, identifying segments like Champions (1636 average PL spend) and Bad (207 average PL spend)

Opportunities in at-risk clusters

We focused on To Reactivate and About to Sleep clusters, which combine high potential (891-1042 PL spend) with churn risk, these clusters represent critical opportunities to retain customers through private label strategies







6.1 PRIVATE LABELS IN LOYALTY MECHANISMS: ANALYSIS AND INSIGHTS

PL AS LOYALTY DRIVER

LESSON LEARNED

Non-churners consistently spend more on PLs, and their higher PL contribution to total sales further highlights the importance of PL in customer loyalty. Indeed, Non-churners spend on PL 870 vs. 324 for churners. **PL products also contribute more** to non-churners' total sales, having a more significant role in retaining customers, indicating that they could be an effective tool for loyalty

Customers in premium clusters (Champions, Loyal) spend more than other clusters on PL, underlining their role in sustaining loyalty. **Champions and Loyal** (with also Uknown and Cannot Lose) clusters have higher PL engagement with 1636-1441 PL spend and contribution of 32.1%-33.1%, confirming their value as the most engaged customers

The **Bad cluster's low PL engagement** and significant churn risk infact they have a PL contribution of 26% and a PL engagement of 207

RECCOMANDATIONS

Re-engage At-Risk Clusters

Clusters like To Reactivate and About to Sleep demonstrate significant potential for retention with their above-average engagement in private labels. Targeted campaigns could include personalized vouchers offering discounts on their most purchased private label items or exclusive "limited-time bundles" that appeal to their shopping preferences. Additionally, implementing new and better ADV about new private label launches or promotions could draw these customers back into regular spending patterns, reducing churn risk

Strengthen Loyalty Through Private Labels

Private labels are key to sustaining loyalty in clusters like Champions and Loyal. Introduce tiered rewards, such as bonus points or exclusive access to premium PL products. Programs like a "Private Label Club" offering perks for high spenders could further deepen their connection

Unlock Opportunities in Low-Engagement Segments

The Bad cluster presents growth potential despite low PL engagement. Free PL samples, first-purchase discounts, or cross-promotions with national brands can encourage trials

7. SUM-UP & CONCLUSIONS

RFM

Customers were classified using RFM **to uncover distinct behavior** patterns and prioritize marketing efforts. Key findings show that high-value clusters like Champions and Potential Loyalists contribute significantly to income, whilst segments like Bad and About to Sleep require re-engagement techniques. The **goal is to increase retention, stimulate upselling, and efficiently manage resources across client tiers for long-term growth**

MBA

The Market Basket Analysis revealed that customers favor **quick meal items** (e.g., Pane, Salumi), **healthy produce** (e.g., Banane, Agrumi), and **bulk cleaning products** (e.g., Detergenti Superfici). The analysis used an **iterative approach to uncover key purchasing patterns**, categorizing products as either rule generators (e.g., Banane, Pane) or rule consequences (e.g., Formaggi Freschi), offering insights for inventory and marketing strategies.

CHURN

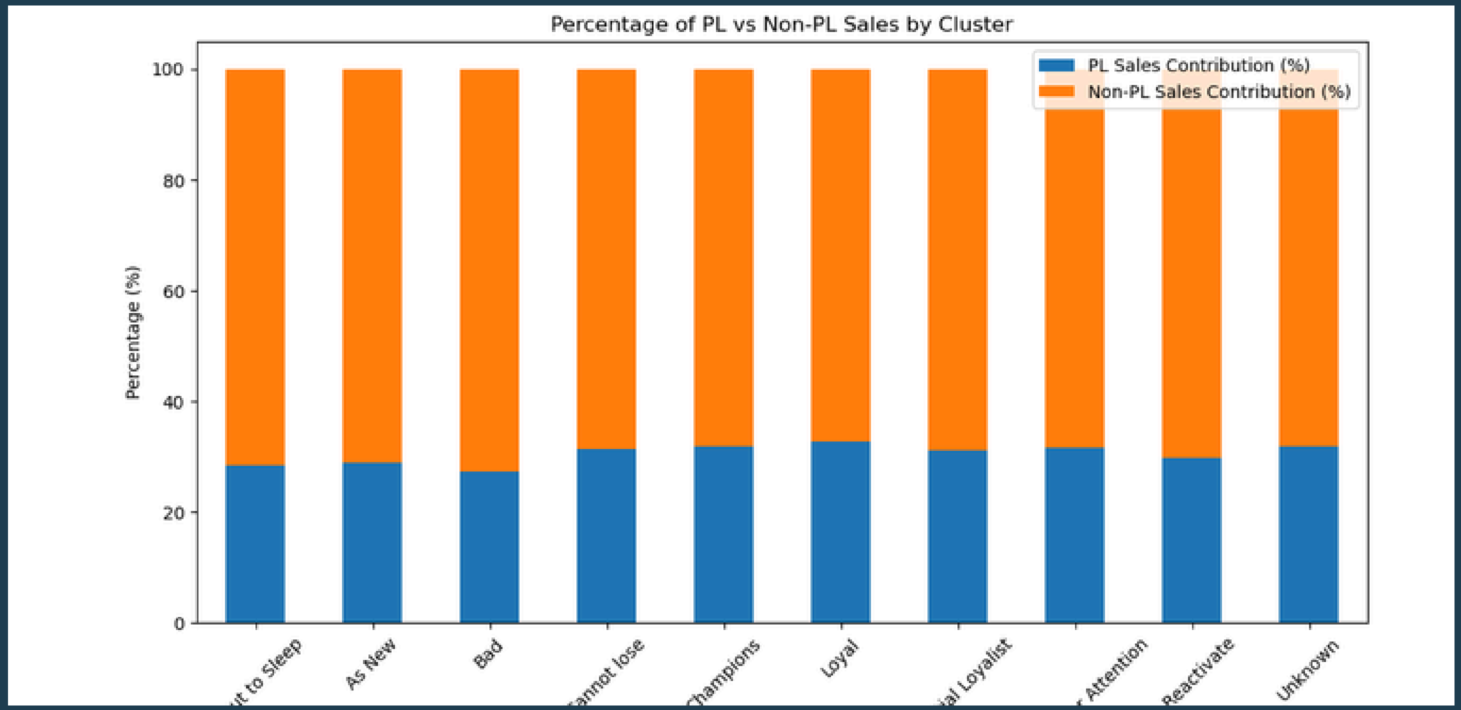
The churn analysis conducted mainly through Logisti Regression, XGBoost and Survival Analysis, has helped identify the key variables to focus on in order to predict customer churn more effectively. While some of these variables are intuitive, such as the fact that **customers who visit infrequently are more likely to churn**, others have revealed that factors like a **low average spend** and, more importantly, **purchasing few private label items**, can also be **indicators of potential churn**.

PL and LOYAL MECHANISM

PL are crucial for **fostering loyalty**, **driving engagement** in high-value clusters like Champions and Loyal, while also **offering retention opportunities** in at-risk segments like To Reactivate and About to Sleep. Leveraging targeted campaigns, personalized incentives, and strategic **PL promotions can enhance loyalty** among top customers and re-engage those at risk, highlighting private labels as a foundation for sustainable loyalty strategies.

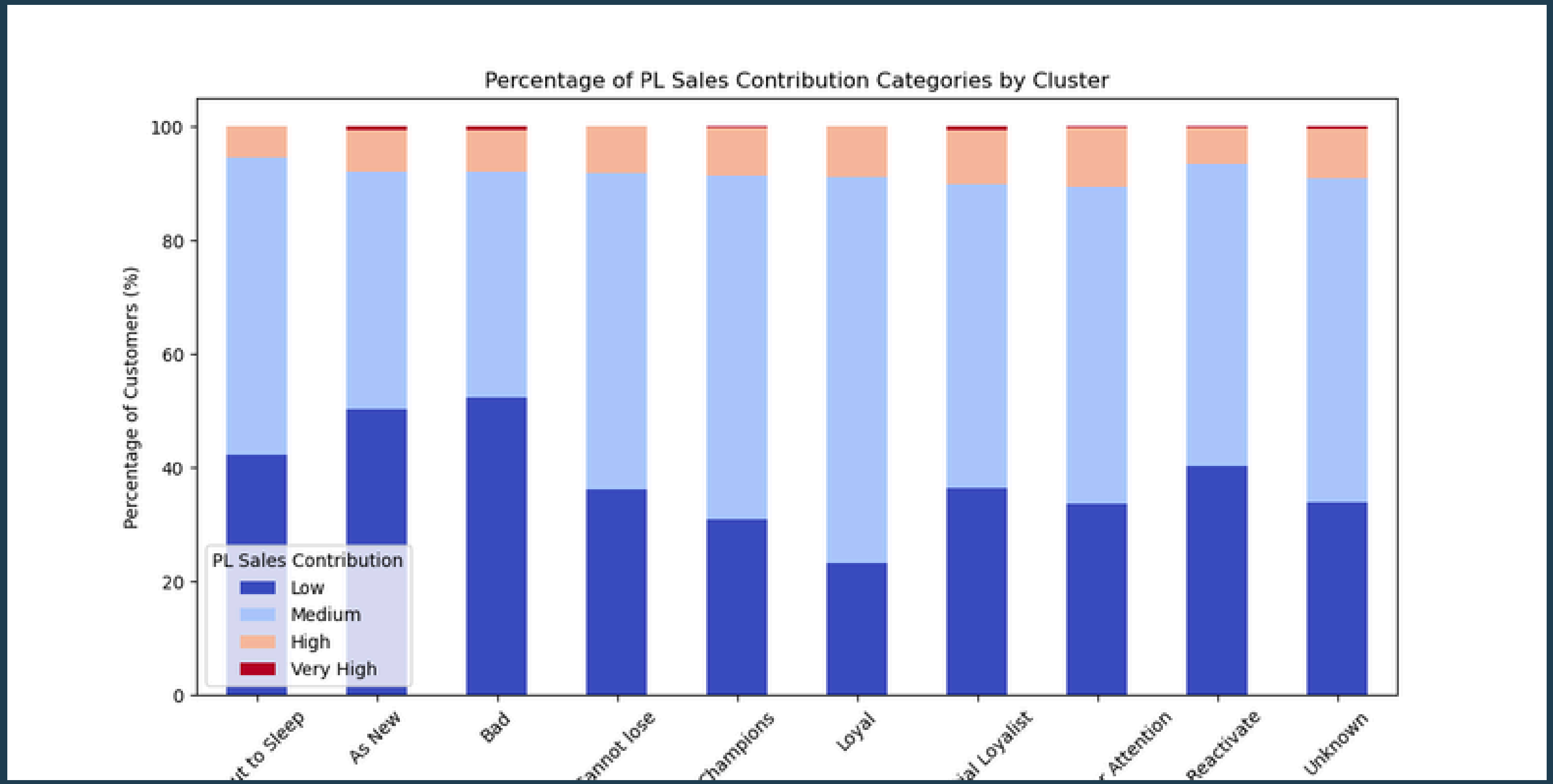
ANNEX - 1. PL EFFECT ON SALES

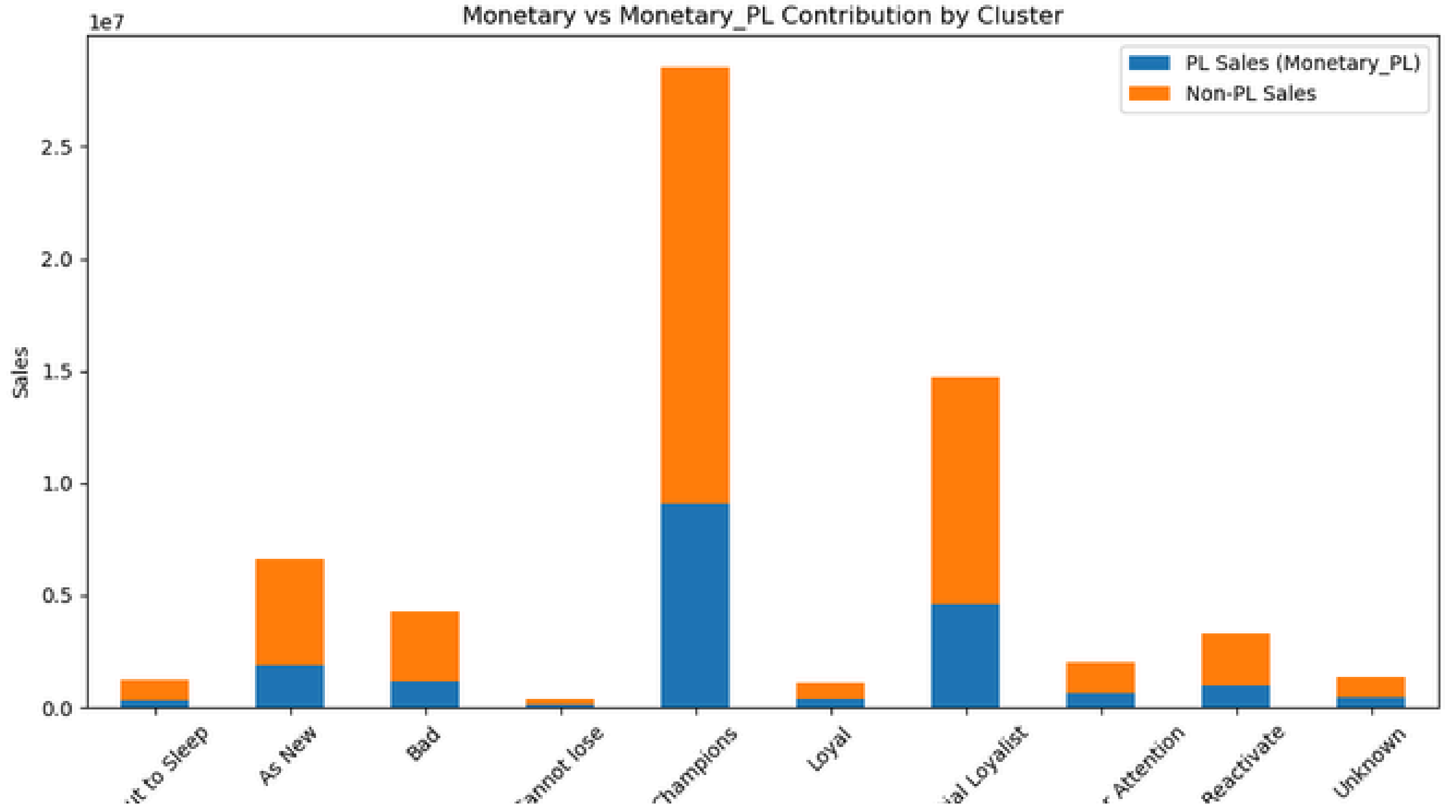
The following charts provide a detailed analysis of the role of private labels in correlation with the RFM clusters.



Clusters such as ***Champions*** and ***Potential Loyalists*** exhibit a higher proportion of Non-PL sales, indicating a **preference for branded products**. For all segments, PL contribution are between 25-35%. Clusters like ***Bad*** and ***As New*** have **relatively higher PL sales contributions**, suggesting that PL products might be more popular in segments with low loyalty or new customers. PL sales contribution is consistent but **generally lower than Non-PL sales in all clusters.**

Also, it can be seen in the graph below that around 5-10% of the customers affect total PL sales highly by contributing PL high.The affect of PL is at medium level for around half of the customers for championship,loyal,unknown, cannot lose and potential loyalist... It is highly low level for cluster bad around 50%





The chart above shows that the monetary values across the segments highlight key revenue contributors. *Champions* generate the highest sales, followed by *Potential Loyalists*, both being critical for revenues. *As New* and *Bad* segments contribute moderately, showing potential for growth with targeted campaigns. .

Champions exhibit the highest overall sales, but the PL contribution remains modest, reinforcing their preference for branded products. *Bad* and *As New* clusters show a more balanced contribution between PL and Non-PL sales, reflecting their reliance on value-oriented PL offerings. Smaller clusters such as *Cannot Lose* and *Loyal* have lower total sales but show varied PL contributions, suggesting niche preferences.